

Integration Formulae – Quick Reference Sheet

Transcribed & reformatted from teacher's notes | ISC Class XII

A. Standard Integrals – Power, Exponential, Log

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c \quad (n \neq -1) \quad 1$$

$$\int dx = x + c \quad 2$$

$$\int \frac{1}{x} dx = \log |x| + c \quad 3$$

$$\int a^x dx = \frac{a^x}{\log a} + c \quad 4$$

$$\int e^x dx = e^x + c \quad 5$$

$$\int \sqrt{x} dx = \frac{2}{3} x^{3/2} + c \quad 12$$

$$\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + c \quad 13$$

B. Trigonometric Integrals

$$\int \sin x dx = -\cos x + c \quad 6$$

$$\int \cos x dx = \sin x + c \quad 7$$

$$\int \sec^2 x dx = \tan x + c \quad 8$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + c \quad 9$$

$$\int \sec x \tan x dx = \sec x + c \quad 10$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c \quad 11$$

$$\int \tan x dx = -\log |\cos x| + c = \log |\sec x| + c \quad 16$$

$$\int \cot x dx = \log |\sin x| + c \quad 17$$

$$\int \sec x dx = \log |\sec x + \tan x| + c = \log \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + c \quad 18$$

$$\int \operatorname{cosec} x dx = \log |\operatorname{cosec} x - \cot x| + c = \log \left| \tan \frac{x}{2} \right| + c \quad 19$$

C. Chain-Rule Type $(f(x), f'(x))$ Forms

$$\int f^n(x) f'(x) dx = \frac{f^{n+1}(x)}{n+1} + c \quad 14$$

$$\int \frac{f'(x)}{f(x)} dx = \log |f(x)| + c \quad 15$$

$$\int e^x (f(x) + f'(x)) dx = e^x f(x) + c \quad 31$$

D. Integrals Leading to Inverse Trig / Log Forms

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c \quad 20$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c \quad 21$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + c \quad 22$$

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c \quad 23$$

$$\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1} \left(\frac{x}{a} \right) + c \quad 24$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \log \left| x + \sqrt{x^2 + a^2} \right| + c \quad 25$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log \left| x + \sqrt{x^2 - a^2} \right| + c \quad 26$$

E. Integrals of $\sqrt{a^2 \pm x^2}$ Type

$$\int \sqrt{a^2 - x^2} dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + c \quad 27$$

$$\int \sqrt{x^2 + a^2} dx = \frac{x\sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \log \left| x + \sqrt{x^2 + a^2} \right| + c \quad 28$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + c \quad 29$$

F. Integration by Parts

$$\int uv dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx \quad 30$$

Choosing u – the ILATE rule

When both factors could be u , pick u in this priority order (the one appearing *first* in the list is taken as u):

I	Inverse trig.	$\sin^{-1} x, \tan^{-1} x, \dots$
L	Logarithmic	$\log x$
A	Algebraic	$x, x^2, x^3 + 1, \dots$
T	Trigonometric	$\sin x, \cos x, \dots$
E	Exponential	e^x, a^x

G. Properties of Definite Integrals

$$\int_a^b f(x) dx = - \int_b^a f(x) dx \quad 32$$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx \quad 33$$

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx \quad 34$$

$$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx \quad 35$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f \text{ even; } = 0 \text{ if } f \text{ odd} \quad 36$$

Tip – Completing the Square (Rule 37)

Add *and* subtract $\left(\frac{b}{2a}\right)^2$ inside the bracket of $ax^2 + bx + c$ – this becomes the third term of the perfect square:

$$\begin{aligned} ax^2 + bx + c &= a \left[x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 \right] + c \\ &= a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right) \end{aligned}$$

H. Weierstrass Substitution (worked pattern)

For $\int \frac{dx}{a + b \cos x}$ or $\int \frac{dx}{a + b \sin x}$, let $\tan \frac{x}{2} = t$. Then:

$$\begin{aligned} \sec^2 \frac{x}{2} \cdot \frac{1}{2} dx &= dt \implies dx = \frac{2 dt}{1 + t^2} \\ \cos x &= \frac{1 - t^2}{1 + t^2}, \quad \sin x = \frac{2t}{1 + t^2} \end{aligned}$$

Substitute all three before simplifying. Always back-substitute $t = \tan \frac{x}{2}$ in the final answer.

Quick Recognition Guide

$f(x)f'(x)$ form present	Rule C1
$\frac{f'(x)}{f(x)}$ present	Rule C2 ($\log f(x) $)
$e^x[f(x) + f'(x)]$ present	Rule C3
Product of two unlike functions	Parts (F) – ILATE
$\frac{1}{a \pm b \cos x}$ or $\frac{1}{a \pm b \sin x}$	Weierstrass $t = \tan \frac{x}{2}$ (H)
$\sqrt{a^2 - x^2}, \sqrt{x^2 \pm a^2}$ alone	Direct results (E)
Quadratic under root/denominator	Complete square (37), then D/E